attaching iron drills to a tractor. Seed drills ensure that the seeds are planted at equal intervals and at the correct depth in the soil.



**Figure 21.5** Seed Drill

**c. Dibbling**

It is the placement of seed material in a furrow, pit or hole at predetermined spacing with a dibble, more commonly by hand. Soil around the hole is pressed with hand or leg for moist soil contact.



**Figure 21.6** Dibbling

## More to know

Transplanting is removal of an actively growing seedling from one place (usually nursery bed) and planting it in the main field for further growth till harvest. Transplanting makes use of pre-grown plants, seedlings or vegetative propagated clones.



## 21.2.3 Adding Manure and Fertilisers

The substances which are added to the soil in the form of nutrients to enhance the growth of plants are called **manure** and **fertilisers**. The term fertility refers to the inherent capacity of a soil to supply nutrients to crop plants in adequate amounts and in suitable proportions. These nutrients are essential for the growth of plants.

Manure is an organic substance obtained from the decomposition of plants or animal wastes. Farmers dump plant and animal waste in pits at open places and allow it to decompose. The decomposed matter is used as organic manure. Regular addition of organic manures helps to maintain the soil fertility, protecting them from wind and water erosion and preventing nutrient losses through runoff and leaching. This also increases water-holding capacity, soil aggregation, soil aeration and permeability.

### Activity 2

Set up a compost pit within your school compound. Put all the organic wastes like food waste and plant leaf in your school campus, cover it with soil. Wait for three weeks and then you can use this as manure for the plants in your school.

Fertilizer is a substance which is added to the soil to improve plants' growth and yield. Fertilizers are composed mainly of Urea, Ammonium sulphate, Super phosphate, Potash and NPK (Nitrogen, Phosphorus, Potassium). The use of synthetic fertilizers has significantly improved the quantity of the food available today, although their long-term use is debated by environmentalists.



Manure Fertilizer

**Figure 21.7** Manure and Fertilizer



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